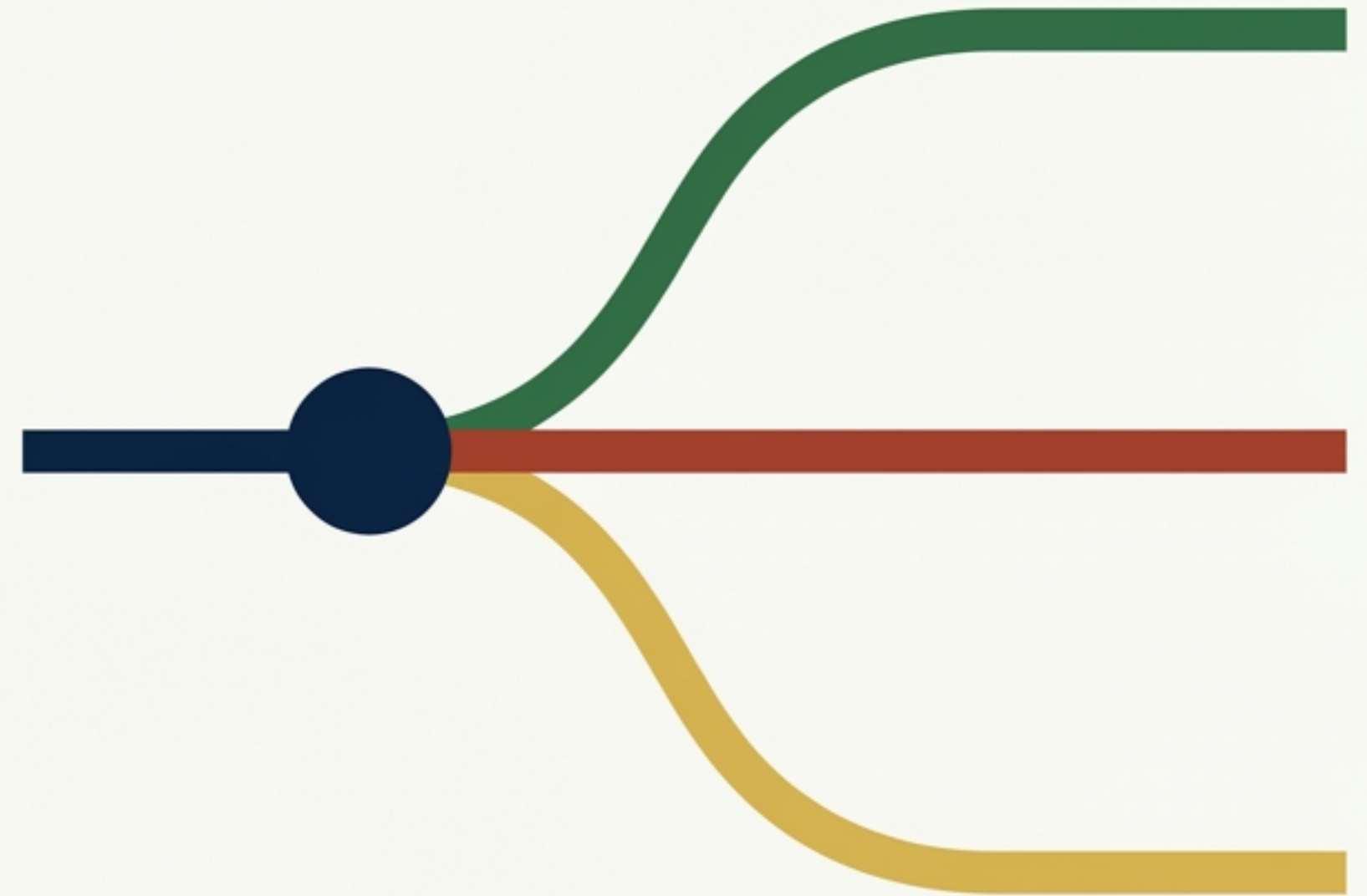


Softmax Regression

Scaling Logistic Regression
for Multi-Class Problems



The Binary Limitation

Standard logistic regression is built to answer "Yes or No". When real-world data introduces three or more potential outcomes, the standard model architecture fails.

Binary Classification

CGPA	IQ	Placement
7.1	85	Yes
8.5	95	Yes
6.2	70	No

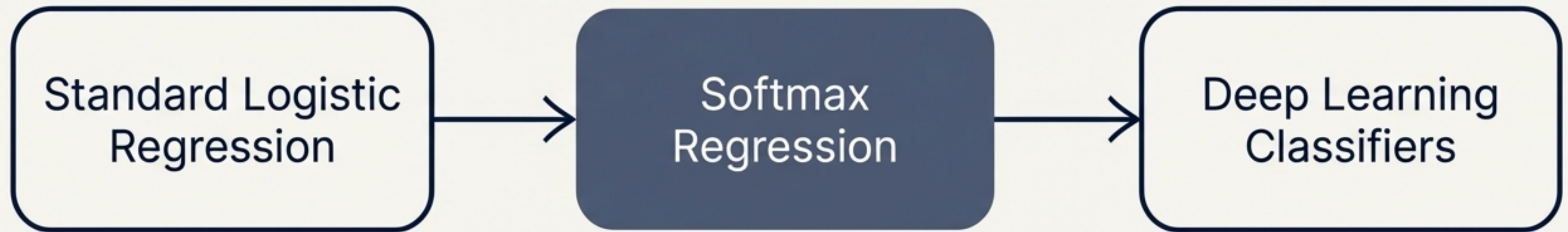


Real-World Classification

CGPA	IQ	Placement
7.1	85	Yes
8.5	95	Opt-Out
6.2	70	No

Enter Softmax (Multinomial Logistic Regression)

Softmax Regression is the generalisation of logistic regression for multi-class problems.



Backwards-Compatible: If you set the number of classes to two, a Softmax model behaves exactly like standard binary logistic regression.

The Mathematical Engine: The Softmax Function

Unlike the Sigmoid function which outputs a single probability, Softmax converts a vector of raw scores into an array of probabilities for every possible class.

$$\sigma(Z_i) = \frac{e^{Z_i}}{\sum_{k=1}^K e^{Z_k}}$$

Target Class Score (exponentiated to ensure positive values) ————— e^{Z_i}

Normalisation Term (sum of all exponentiated class scores) ————— $\sum_{k=1}^K e^{Z_k}$

K ————— Total number of classes

The Golden Rule of Softmax

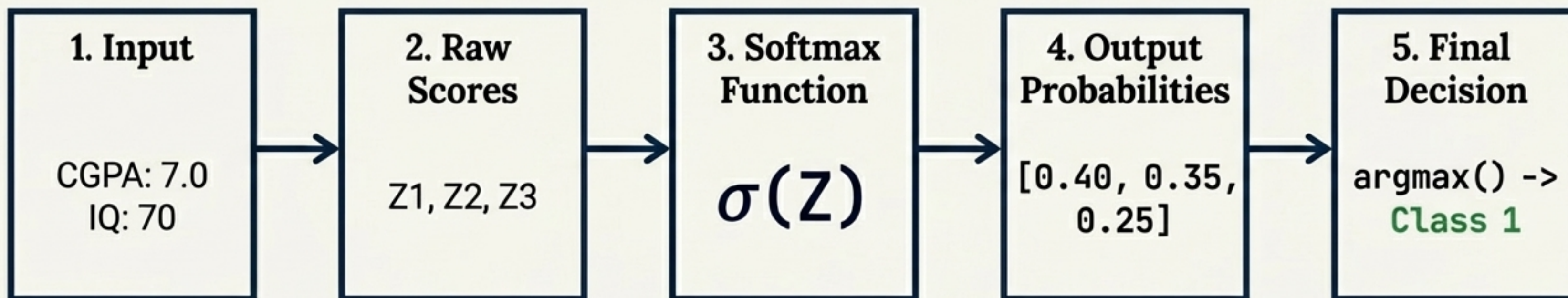
Softmax guarantees that the output value for every class will strictly fall between 0 and 1. Crucially, the sum of the probabilities across all classes will always equal exactly 1.



$$0.50 + 0.25 + 0.25 = 1.0$$

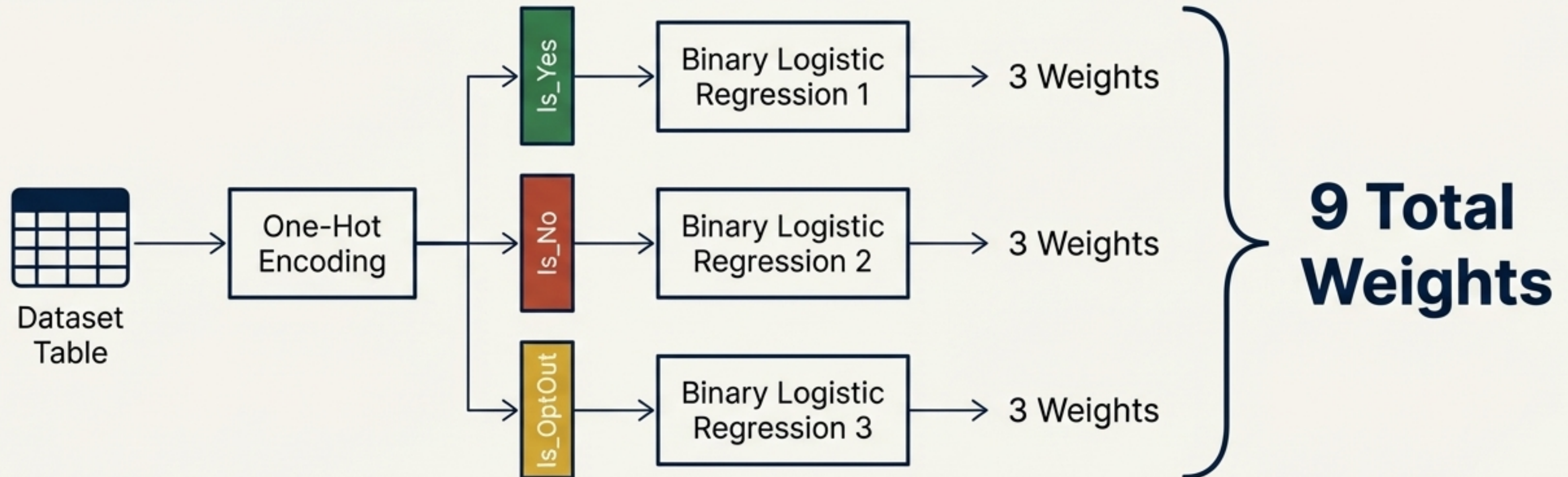
Prediction Intuition: Making a Choice

The model calculates an independent score for every class, normalises them into probabilities, and assigns the data point to the class with the maximum probability.



Training Intuition: The Multiple Models Mental Model

Conceptually, training a multi-class model is like isolating each class and building a dedicated binary classifier for it. For 3 classes with 2 input features, this theoretical approach yields 3 separate models, each calculating 3 weights for a total of 9 weights.



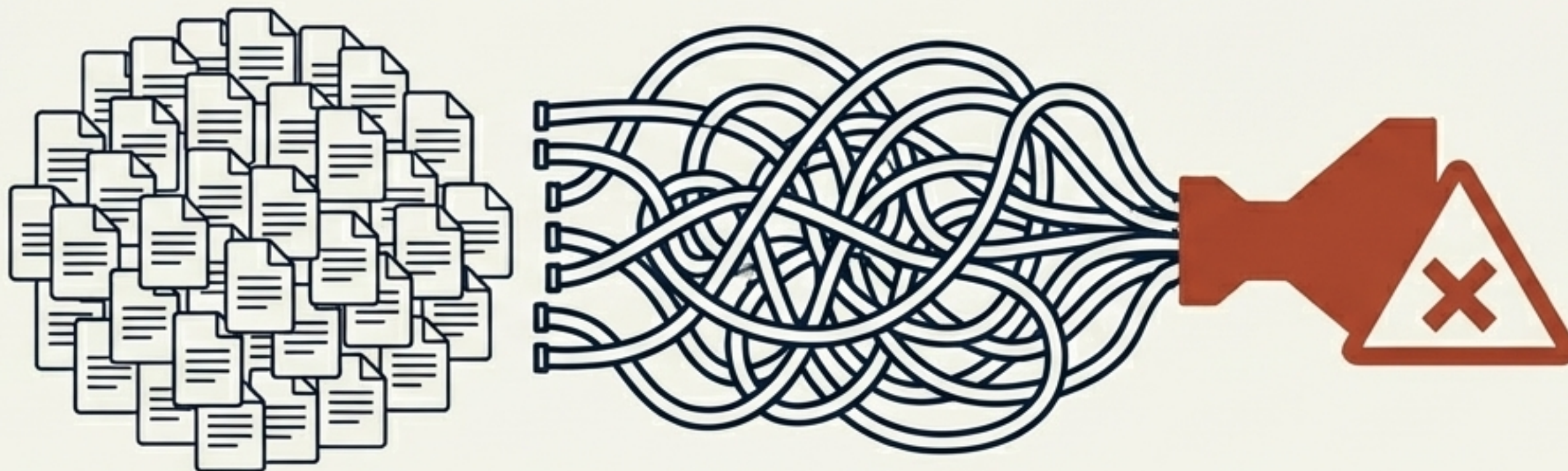
The Reality of Training

Training K separate logistic regression models works for small datasets, but fails at scale. Training individual models sequentially is computationally expensive and highly inefficient. The solution requires unifying the architecture.

Small Scale

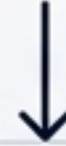


Enterprise Scale



Modifying the Loss Function

Instead of running multiple models, we alter the loss function itself. By replacing standard binary log-loss with **Categorical Cross-Entropy**, a single algorithm can optimise weights for all classes simultaneously.



$$L = - \sum_{i=1}^M \sum_{k=1}^K Y_{ik} \log(Y_{\text{hat}}_{ik})$$

Simplifying the Math

Because the actual target is one-hot encoded, only the true class has a value of 1. All other classes have a value of 0. This instantly zeros out the non-target terms, collapsing the intimidating double-summation into a simple, efficient calculation.

$$1 \times \log(Y_hat_1) \checkmark$$

$$~~0 \times \log(Y_hat_2)~~$$

$$~~0 \times \log(Y_hat_3)~~$$

Simultaneous Gradient Descent

Using the unified loss function, gradient descent calculates the derivatives and updates all 9 weights simultaneously in a single matrix operation. One model, one training loop, all weights optimised.

Gradient Descent Update


$$\begin{bmatrix} W_{11} & W_{12} & W_{10} \\ W_{21} & W_{22} & W_{20} \\ W_{31} & W_{32} & W_{30} \end{bmatrix}$$
$$W_{\text{new}} = W_{\text{old}} - \eta \left(\frac{\partial L}{\partial W} \right)$$

Practical Implementation: Scikit-Learn

In scikit-learn, Softmax is natively supported. By passing the `multi_class='multinomial'` parameter, the standard LogisticRegression class immediately upgrades from a binary classifier to a multi-class Softmax engine.

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(multi_class='multinomial')
model.fit(X_train, y_train)
```

Under the Hood of the Code

Calling `predict_proba()` reveals the Softmax function at work. The model outputs a continuous probability array across all three classes, and just like the formula dictates, they sum exactly to 1.

```
model.predict_proba(new_data)
```

```
[[ 8.0e-04,  0.72,  0.27 ]]
```

Setosa

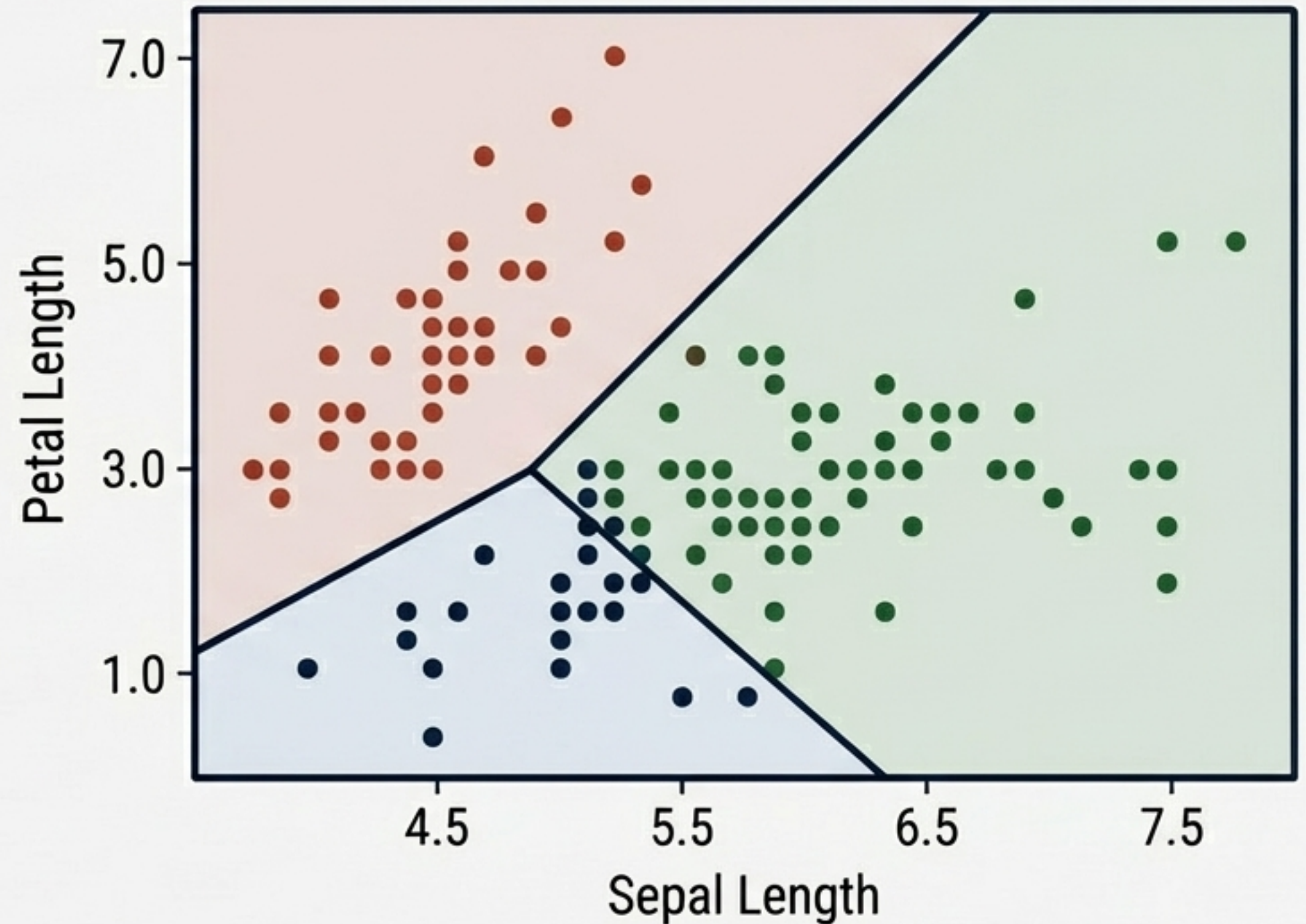
Versicolor

Virginica

Sum = 1.0

Visualising Decision Boundaries

Standard binary regression draws a single line. Softmax regression carves the feature space into distinct, intersecting regions, allowing for complex, multi-territory classification.



Summary: Binary vs. Multinomial

Softmax regression retains the elegance of logistic regression while dramatically expanding its architectural capacity.

	Binary Logistic Regression	Softmax Regression
Classes	2	>2
Core Function	Sigmoid	Softmax
Loss Function	Binary Log Loss	Categorical Cross-Entropy
Outputs	1 Probability	K Probabilities (Summing to 1)
Boundaries	Single Line	Multiple Intersecting Regions